



**Vertical
Institute**

Data Science Bootcamp Written Assessment

SECTION A

Q1: What is Jupyter Notebook and how is it used in data science?

Jupyter Notebook is a tool where you can write and run Python code in your browser. You can run the code in small parts (cells) and see the results straight away, like tables and charts. In data science we use it to explore data, clean it, make graphs and build models.

Q2 Name the steps in a data science life cycle

1. Define the problem
2. Collect the data
3. Clean the data
4. Explore the data (EDA)
5. Build the model
6. Evaluate the model
7. Deploy the model

SECTION B

Q1: How do you reverse a string in Python?

You can use slicing with `[::-1]`.

Example: `s = "hello"` then `s[::-1]` gives "olleh"

Q2: How do you access the last element of a list in Python?

You use index `-1`.

Example: `my_list = [1, 2, 3]` then `my_list[-1]` gives 3

SECTION C

Q1: How do you import a module in Python?

You use the import keyword.

Example: `import pandas as pd` or `from math import sqrt`

Q2: What is a NumPy array?

A NumPy array is like a Python list but faster and better for maths. All the items in it must be the same type. You can do calculations on the whole array at once.

Example: `np.array([1, 2, 3])`

SECTION D

Q1: What is the purpose of exploratory data analysis?

EDA is used to understand the data before building a model. We look at the data, check for missing values and outliers, and use charts and statistics to find patterns and relationships between the columns.

Q2: What is matplotlib, and how is it used in data science?

Matplotlib is a Python library for making charts and graphs. In data science we use it to visualise the data, for example line charts, bar charts, histograms and scatter plots, so it is easier to see patterns.

SECTION E

Q1: What is a correlation?

Correlation shows how strongly two variables are related. It goes from -1 to 1. 1 means when one goes up the other also goes up, -1 means when one goes up the other goes down, and 0 means no relationship.

Q2: What is linear regression?

Linear regression is a machine learning model that predicts a number, like a price. It finds the best straight line through the data to show the relationship between the inputs (X) and the output (y).

Example: predicting car price using horsepower and engine size.

SECTION F

Q1: What is classification, and how is it performed in Python?

Classification is when the model predicts which category something belongs to, for example yes or no. In Python we use scikit-learn: split the data into train and test, choose a model like Decision Tree, Logistic Regression or KNN, use `.fit()` to train and `.predict()` to make predictions, then check the accuracy.

Q2: What is a classification problem?

A classification problem is a problem where the answer is a category, not a number. For example, is an email spam or not spam, or is a tumour malignant or benign.

SECTION G

Q1: What is the purpose of hyperparameter tuning?

Hyperparameters are settings of the model that we choose before training, like `max_depth` for a decision tree. Hyperparameter tuning is trying different settings to find the ones that give the best results. We can use `GridSearchCV` to do this.

Q2: What is a confusion matrix?

A confusion matrix is a table that shows how many predictions the classification model got right and wrong. It shows true positives, true negatives, false positives and false negatives. It helps us see what kind of mistakes the model makes.